

BLUESTOCK MUTUAL FUND ANALYTICS CAPSTONE PROJECT

Final Project Report

Submitted By:

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Project Duration:

Day 1 – Day 7

Technologies Used:

Python, Pandas, NumPy, SQLite, SQL Alchemy, Matplotlib, Seaborn, Plotly, Power BI, GitHub.

Executive Summary:

The Mutual Fund Analytics Capstone Project was developed to build a comprehensive analytics platform capable of processing, analysing, and visualizing mutual fund industry data. The project integrates data engineering, database management, financial analytics, risk analysis, and business intelligence into a unified workflow.

The project utilized ten datasets containing information on mutual fund schemes, Net Asset Value (NAV) history, Assets Under Management (AUM), investor transactions, SIP inflows, category inflows, portfolio holdings, and benchmark indices. These datasets were cleaned, validated, and transformed into analytical datasets suitable for financial analysis and dashboard reporting.

A complete Extract-Transform-Load (ETL) pipeline was implemented using Python. Cleaned data was stored in a SQLite database designed using a star schema architecture. Exploratory Data Analysis (EDA) was performed to identify industry trends, investor behaviour, geographic patterns, category preferences, and sector allocations.

Advanced financial metrics such as CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, Value at Risk (VAR), Conditional Value at Risk (CVaR), Rolling Sharpe Ratio, Investor Cohort Analysis, SIP Continuity Analysis, and Portfolio Concentration Analysis were computed. A rule-based fund recommendation system was also developed to recommend suitable funds based on investor risk profiles.

The final solution includes an interactive Power BI dashboard with multiple analytical pages enabling users to explore mutual fund performance, investor activity, and market trends. The project demonstrates the practical application of Python, SQL, and Power BI in the field of financial analytics.

1. Project Objective:

The primary objective of this project was to develop an end-to-end Mutual Fund Analytics platform capable of:

- Ingesting and validating mutual fund data from multiple sources.
 - Cleaning and transforming raw data into analytical datasets.
 - Designing and implementing a relational database structure.
 - Performing exploratory and advanced analytics on mutual fund data.
 - Evaluating fund performance using industry-standard financial metrics.
 - Assessing downside risk using modern risk management techniques.
 - Building an interactive dashboard for business users.
 - Providing actionable insights for investors and fund managers.
-

2. Data Sources:

The project uses ten datasets representing different aspects of the mutual fund ecosystem.

2.1 Fund Master Dataset

Contains information about mutual fund schemes including AMFI codes, scheme names, fund houses, categories, sub-categories, and risk grades.

2.2 NAV History Dataset

Contains historical Net Asset Value records for mutual fund schemes. This dataset is the primary source for return calculations and risk analysis.

2.3 AUM by Fund House Dataset

Contains Assets Under Management values for different Asset Management Companies (AMCs) across multiple years.

2.4 Monthly SIP Inflows Dataset

Contains monthly Systematic Investment Plan inflows across the mutual fund industry.

2.5 Category Inflows Dataset

Contains category-level investment inflows and outflows.

2.6 Industry Folio Count Dataset

Contains investor folio growth statistics across the study period.

2.7 Scheme Performance Dataset

Contains return percentages, expense ratios, and scheme performance indicators.

2.8 Investor Transactions Dataset

Contains SIP, lumpsum, and redemption transactions with demographic information.

2.9 Portfolio Holdings Dataset

Contains portfolio allocations across sectors and securities.

2.10 Benchmark Indices Dataset

Contains benchmark index performance data used for Alpha, Beta, and tracking error calculations.

3. ETL Design:

The ETL pipeline was developed using Python and Pandas.

Extraction Phase

Raw CSV files were loaded from the data/raw directory. Additional live NAV data was fetched using the MFAPI mutual fund API.

The following datasets were successfully ingested:

- Fund Master
- NAV History
- AUM Data
- SIP Inflows
- Category Inflows
- Industry Folio Counts
- Scheme Performance
- Investor Transactions
- Portfolio Holdings
- Benchmark Indices

Transformation Phase

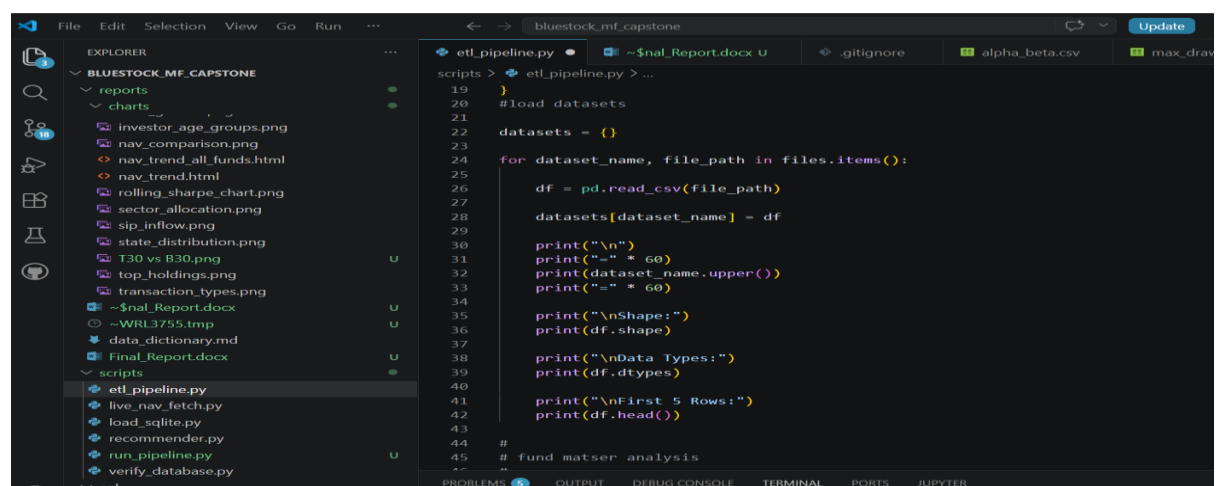
Several data quality checks and transformations were performed:

- Date parsing and standardization
- Duplicate removal
- Missing value handling
- Forward filling of NAV values
- Validation of transaction amounts
- Standardization of transaction types
- Expense ratio validation
- Numeric data type enforcement

Loading Phase

Cleaned datasets were loaded into a SQLite database using SQL Alchemy. The database served as the analytical layer for reporting and dashboard development.

The ETL process ensured consistency, reliability, and reproducibility across the project.



The screenshot displays a Jupyter Notebook environment with a dark theme. On the left, the 'EXPLORER' sidebar shows a file tree for 'BLUESTOCK_MF_CAPSTONE', including folders for 'reports' and 'charts', and various data files like 'investor_age_groups.png', 'nav_comparison.png', 'nav_trend_all_funds.html', 'nav_trend.html', 'rolling_sharpe_chart.png', 'sector_allocation.png', 'sip_inflow.png', 'state_distribution.png', 'T30 vs B30.png', 'top_holdings.png', and 'transaction_types.png'. Below these are report files like '\$nal_Report.docx' and 'Final_Report.docx', and a 'scripts' folder containing 'etl_pipeline.py', 'live_nav_fetch.py', 'load_sqlite.py', 'recommender.py', 'run_pipeline.py', and 'verify_database.py'. The main area shows the 'etl_pipeline.py' script with the following code:

```
19 }
20 #load datasets
21 datasets = {}
22
23 for dataset_name, file_path in files.items():
24
25     df = pd.read_csv(file_path)
26
27     datasets[dataset_name] = df
28
29     print("\n")
30     print("-" * 60)
31     print(dataset_name.upper())
32     print("-" * 60)
33
34     print("\nShape:")
35     print(df.shape)
36
37     print("\nData Types:")
38     print(df.dtypes)
39
40     print("\nFirst 5 Rows:")
41     print(df.head())
42
43
44 # fund master analysis
45 ..
```

4. Database Design:

A SQLite database was designed to support analytical processing and dashboard reporting. A star schema approach was adopted to improve query performance and simplify relationships between datasets.

Dimension Tables

Dim_fund

Stores fund-level information including:

- AMFI Code
- Scheme Name
- Fund House
- Category
- Sub-Category
- Risk Grade

Dim_date

Stores calendar information including:

- Date
- Month
- Quarter
- Year

Fact Tables

Fact_nav

Stores historical NAV records for all mutual fund schemes.

Fact_transactions

Stores investor transaction details such as SIPs, lumpsum investments, and redemptions.

Fact_performance

Stores performance-related metrics including returns and expense ratios.

Fact_aum

Stores Assets Under Management information.

Benefits of Database Design

The database design provides:

- Efficient analytical querying
- Data consistency
- Reduced redundancy
- Improved dashboard performance
- Easy integration with Power BI

A total of ten analytical SQL queries were developed to support reporting and business intelligence requirements.

5. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted to understand trends, patterns, relationships, and anomalies within the mutual fund ecosystem.

The analysis included NAV trends, AUM growth, SIP inflows, category inflows, investor demographics, geographic distribution, folio growth, return correlations, and sector allocations.

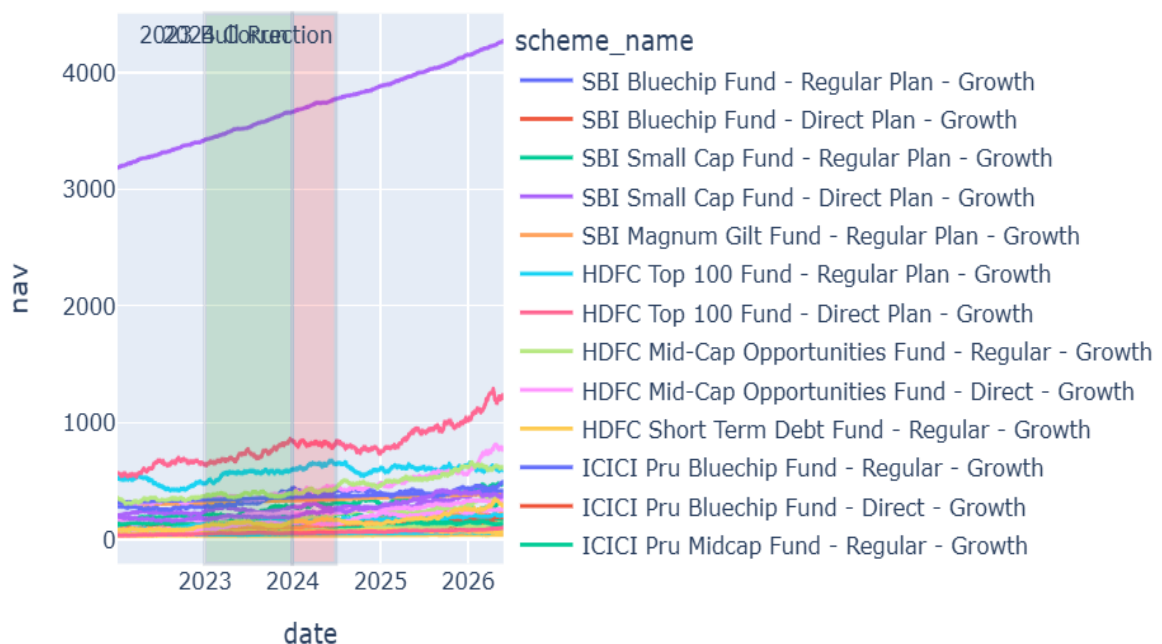
5.1 NAV Trend Analysis:

Historical NAV trends were analysed across all available mutual fund schemes.

Observations:

- Most funds exhibited consistent growth over the study period.
- Market corrections were visible during periods of increased volatility.
- Large-cap funds showed lower volatility compared to mid-cap and sectoral funds.
- Several schemes significantly outperformed industry averages.

Daily NAV Trend (2022-2026)

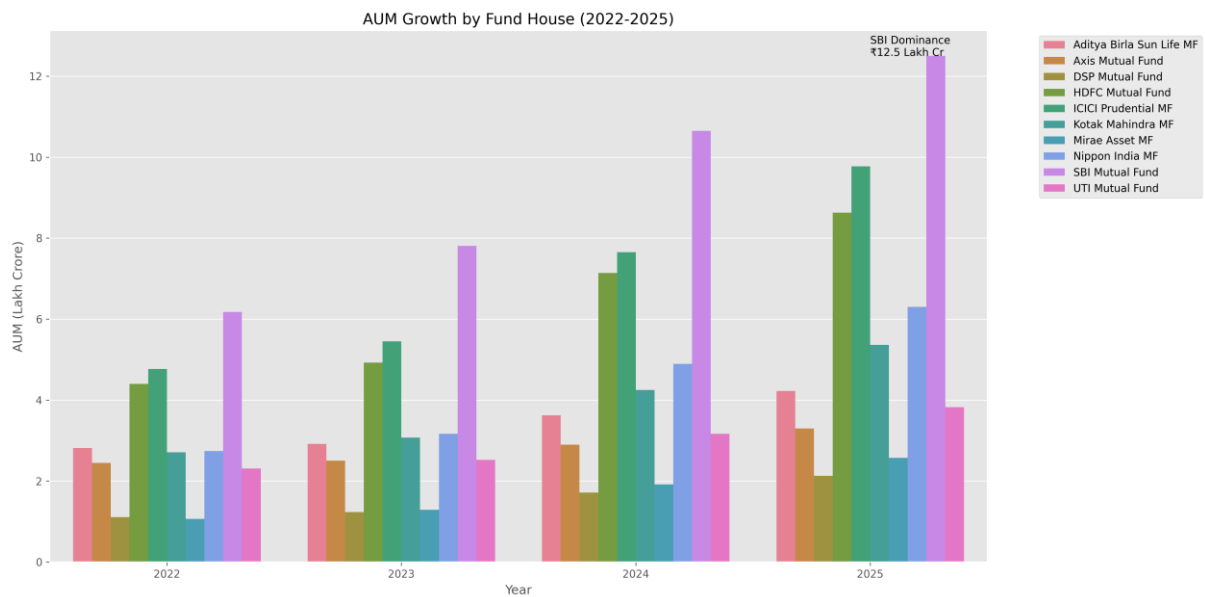


5.2 AUM Growth Analysis:

Assets Under Management (AUM) were analysed across major Asset Management Companies.

Observations:

- Industry AUM increased substantially from 2022 to 2025.
- Large fund houses dominated total market share.
- Consistent investor participation contributed to industry expansion.
- Growth was supported by increasing SIP contributions.

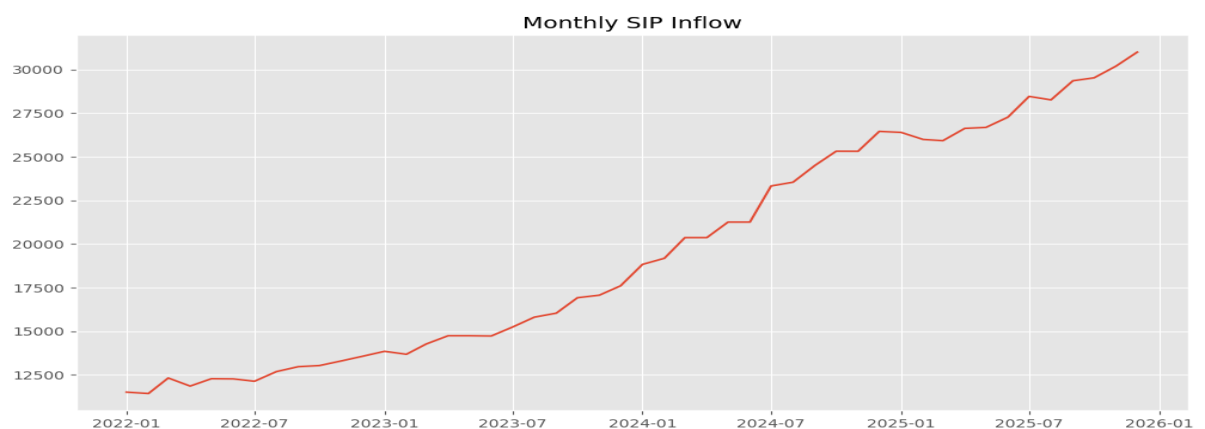


5.3 SIP Inflow Analysis

Monthly SIP inflows were examined to understand investor participation.

Observations:

- SIP inflows displayed a strong upward trend.
- Investor confidence remained strong throughout the study period.
- Record SIP collections were achieved during later years.
- SIP investments became a major contributor to industry growth.

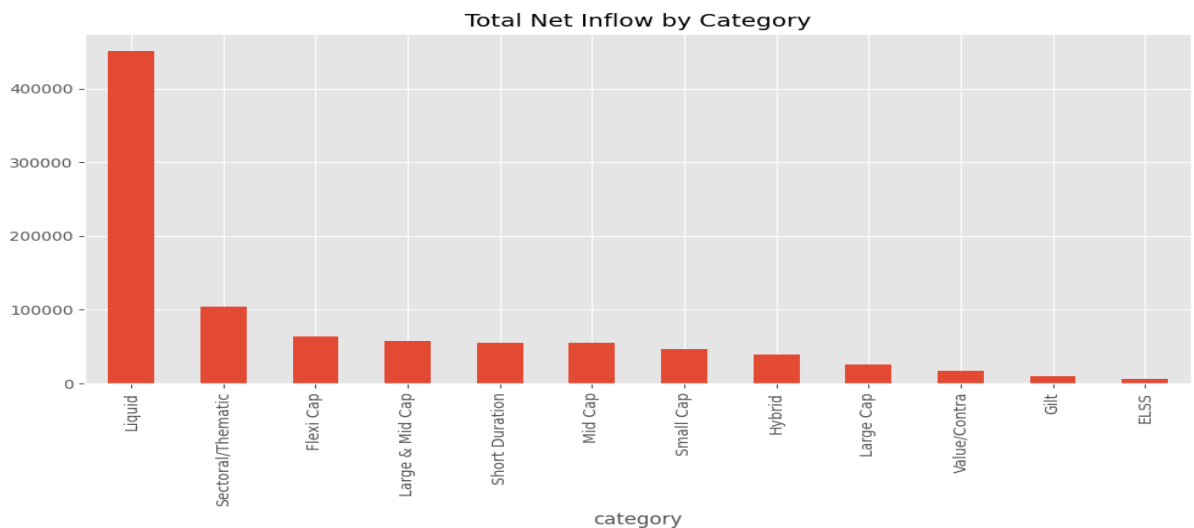


5.4 Category Inflow Analysis:

Category-wise inflows were analysed using heatmaps.

Observations:

- Equity-oriented categories received the highest inflows.
- Investor preference shifted toward growth-oriented investments.
- Certain categories experienced periodic outflows during market uncertainty.
- Category inflows closely followed market sentiment.

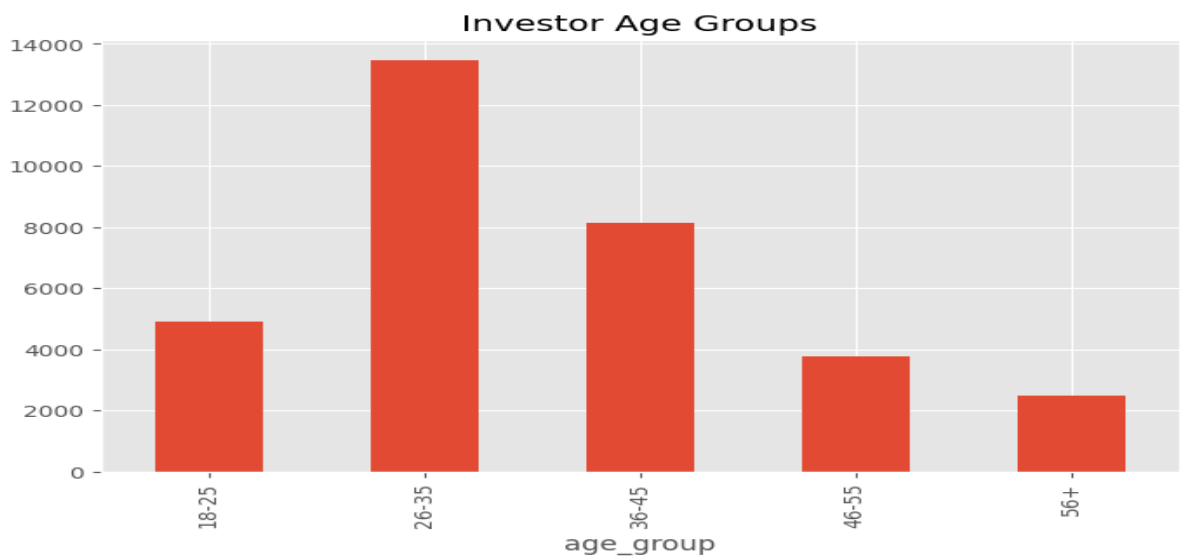


5.5 Investor Demographics Analysis:

Investor age and gender distributions were analysed.

Observations:

- Young investors represented a significant portion of SIP participants.
- Middle-aged investors contributed the highest investment amounts.
- Gender participation showed increasing diversity.
- SIP behaviour varied across demographic groups.

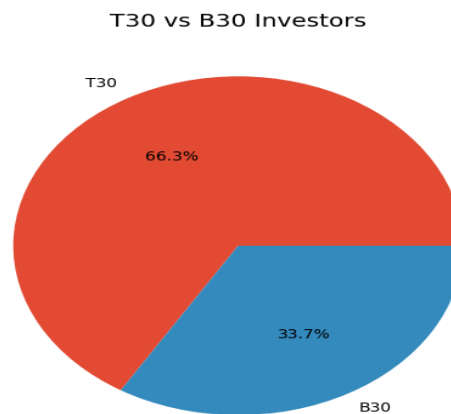


5.6 Geographic Distribution Analysis:

Geographic investment patterns were analysed using state-level and city-tier data.

Observations:

- Major metropolitan regions contributed the largest investment volumes.
- T30 cities dominated total investment activity.
- B30 participation showed gradual improvement.
- Geographic diversification remains an opportunity for industry growth.

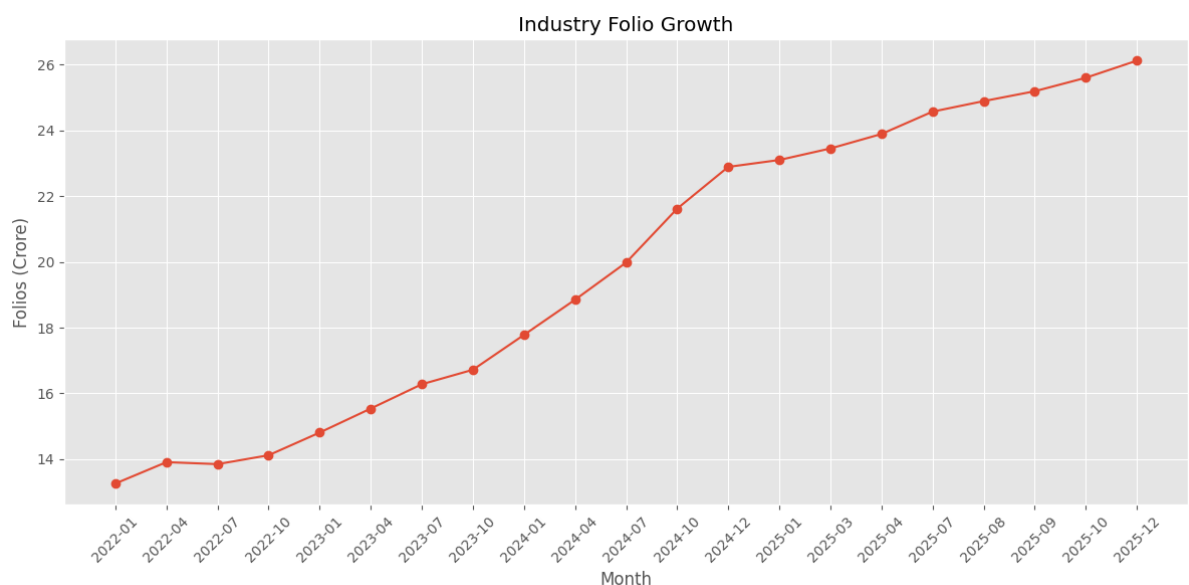


5.7 Folio Growth Analysis:

Investor folio counts were analysed to measure retail participation.

Observations:

- Folio counts increased significantly over the study period.
- Retail participation expanded rapidly.
- Growth indicates increasing awareness of mutual fund investing.
- SIP adoption contributed to folio growth.

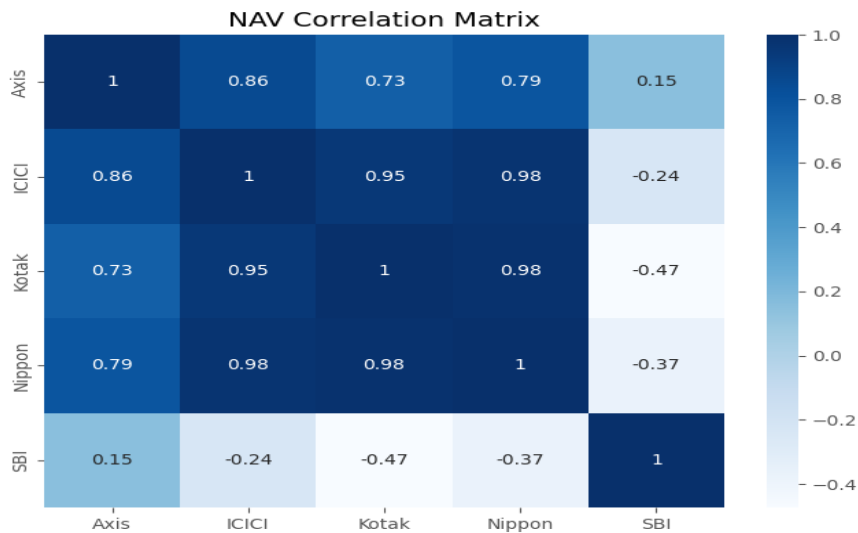


5.8 Correlation Analysis:

Correlation matrices were used to examine relationships between fund returns.

Observations:

- Large-cap funds exhibited strong positive correlations.
- Diversification benefits existed across categories.
- Sector-specific funds displayed unique return characteristics.
- Correlations increased during market stress periods.

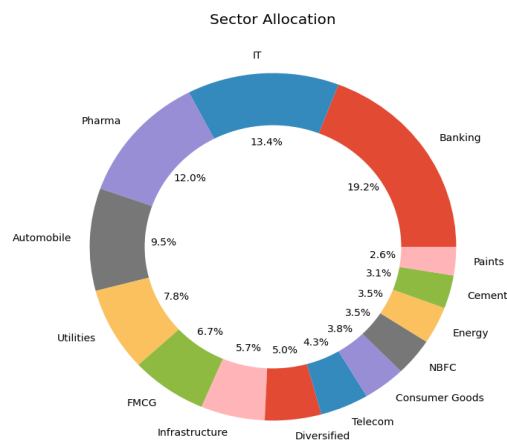


5.9 Sector Allocation Analysis:

Sector allocations were aggregated across equity funds.

Observations:

- Financial services represented the largest sector allocation.
- Technology and consumer sectors also received significant allocations.
- Certain sectors were heavily concentrated among select funds.
- Sector exposure significantly influences portfolio risk.



Key EDA Findings:

1. Mutual fund assets grew consistently throughout the study period.
2. SIP inflows reached record levels, indicating strong investor confidence.
3. Equity categories attracted the highest inflows.
4. Retail investor participation increased substantially.
5. Major metropolitan regions contributed most investments.
6. Large-cap funds exhibited lower volatility.
7. Fund returns showed significant correlations.
8. Sector concentration remained present in multiple portfolios.
9. Folio growth reflected increasing mutual fund adoption.
10. Long-term investing behaviour strengthened over time.

6. Performance Analytics:

Performance analytics was conducted to evaluate mutual fund performance using industry-standard financial metrics. These metrics provide insights into returns, risk-adjusted performance, market sensitivity, and downside risk.

The analysis was performed across all available mutual fund schemes using historical NAV data and benchmark indices.

6.1 Daily Return Analysis:

Daily returns were calculated using the formula:

$$\text{Daily Return} = (\text{NAV Today} / \text{NAV Previous Day}) - 1$$

Daily returns serve as the foundation for all subsequent risk and performance calculations.

Observations:

- Most funds exhibited stable return distributions.
- Market corrections were visible through negative return clusters.
- Large-cap funds generally experienced lower volatility.
- Sector and thematic funds demonstrated higher return variability.

6.2 Compound Annual Growth Rate (CAGR):

CAGR was calculated for 1-Year, 3-Year, and 5-Year periods.

CAGR measures the annualized growth rate of an investment over a specified period.

Benefits:

- Removes short-term market noise.
- Enables comparison between schemes.
- Reflects long-term wealth creation potential.

Observations:

- Several funds achieved strong long-term growth.
- Equity-oriented funds outperformed conservative categories.
- Long investment horizons produced more consistent results.

```

1  amfi_code,CAGR_1Y,CAGR_3Y,CAGR_5Y
2  119551,57.56848841923501,29.18145991652894,
3  119552,5.482448166686016,15.621784755976687,
4  119598,78.61683425540309,25.568431449174977,
5  119599,13.381345873925277,-1.2882996299400817,
6  119120,5.305739305375479,5.6186515164248885,
7  100016,-2.1402903486672065,1.2445994939421334,
8  125497,27.44305456632288,17.69654338000204,
9  100033,50.75653358780363,31.07480621658334,
10 125498,14.812657529198159,10.808126685493423,
11 100025,3.5610696428827415,3.769015761043071,
12 120503,8.152215287087339,13.428793488877178,
13 120504,12.5356431533042,31.11766658791899,
14 120505,28.328204567572858,30.441001673525058,
15 120506,39.94041933796997,17.6845877347511,
16 120507,7.131782819276045,7.111635011762973

```

6.3 Sharpe Ratio Analysis:

The Sharpe Ratio evaluates risk-adjusted performance.

A higher Sharpe Ratio indicates better returns per unit of risk.

Observations:

- Top-ranked funds demonstrated strong risk-adjusted returns.
- Some funds achieved attractive returns but with significantly higher volatility.
- Investors should consider Sharpe Ratio alongside raw returns.

6.4 Sortino Ratio Analysis:

The Sortino Ratio measures downside-risk-adjusted returns.

Unlike Sharpe Ratio, it only considers harmful volatility.

Observations:

- Several funds maintained strong downside protection.
- Funds with high Sortino Ratios generally displayed more consistent performance.
- Sortino Ratio provides a more investor-friendly measure of risk.

```

1  amfi_code,sortino_ratio
2  148567,2.3856443733688186
3  120843,2.3643200321667184
4  148569,2.146913926346055
5  119551,2.140267129061793
6  120505,2.0293529404657464
7  149323,1.8751010639192083
8  118632,1.85013254685726
9  100033,1.8291339615146942
10 120504,1.8052937940992548
11 101206,1.799562874698452
12 119094,1.7037974802463003
13 119598,1.6753165700125519
14 148568,1.632740977873516
15 149324,1.6197931764914093
16 119552,1.609776489628308
17 102885,1.4357420130649492
18 120503,1.3379431500669237

```

6.5 Alpha and Beta Analysis:

Alpha and Beta were calculated using benchmark index returns.

Alpha

Alpha measures excess return beyond benchmark expectations.

Interpretation:

- Positive Alpha = Outperformance
- Negative Alpha = Underperformance

Beta

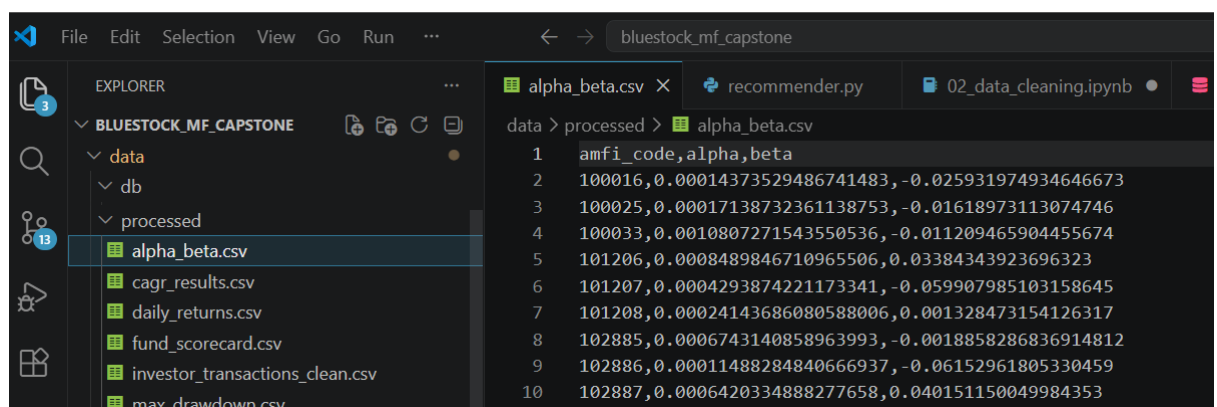
Beta measures sensitivity to market movements.

Interpretation:

- Beta > 1 = More volatile than market
- Beta < 1 = Less volatile than market

Observations:

- Several schemes generated positive Alpha.
- High-Beta funds tended to experience larger market swings.
- Alpha and Beta together provide valuable insights into fund management quality.



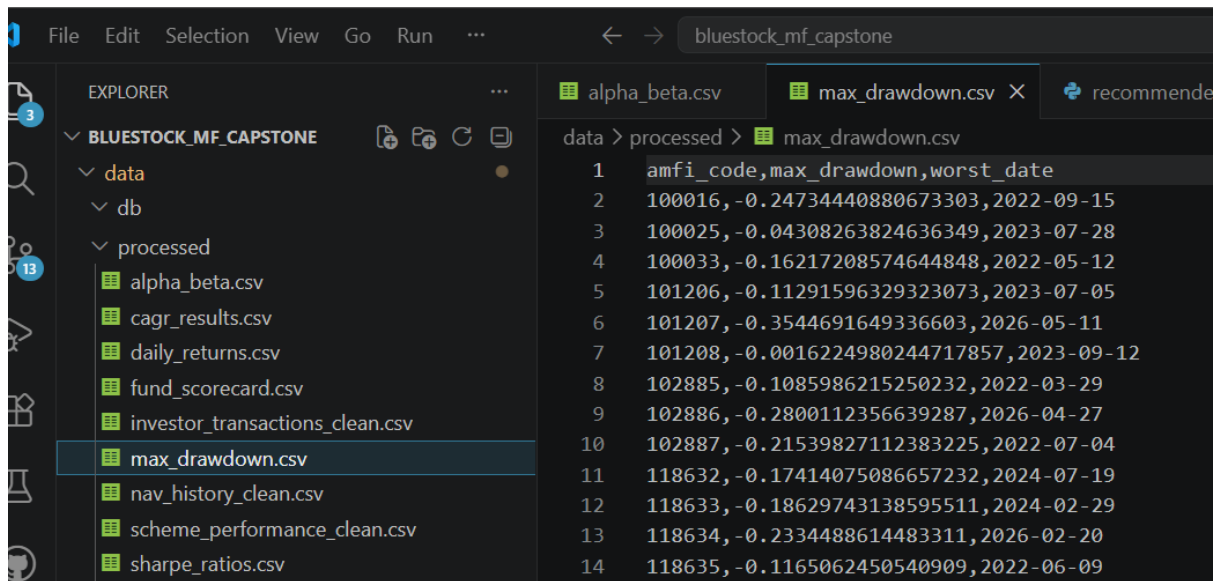
```
data > processed > alpha_beta.csv
1  amfi_code,alpha,beta
2  100016,0.00014373529486741483,-0.025931974934646673
3  100025,0.00017138732361138753,-0.01618973113074746
4  100033,0.0010807271543550536,-0.011209465904455674
5  101206,0.0008489846710965506,0.03384343923696323
6  101207,0.0004293874221173341,-0.059907985103158645
7  101208,0.00024143686080588006,0.001328473154126317
8  102885,0.0006743140858963993,-0.0018858286836914812
9  102886,0.00011488284840666937,-0.06152961805330459
10 102887,0.0006420334888277658,0.040151150049984353
```

6.6 Maximum Drawdown Analysis:

Maximum Drawdown measures the largest peak-to-trough decline experienced by a fund.

Observations:

- Drawdowns varied significantly across schemes.
- High-growth funds often experienced larger drawdowns.
- Maximum Drawdown highlights downside risk that investors may encounter.



6.7 Fund Scorecard:

A composite scoring framework was developed to rank funds.

The scorecard combines:

- 30% CAGR Ranking
- 25% Sharpe Ratio Ranking
- 20% Alpha Ranking
- 15% Expense Ratio Ranking
- 10% Maximum Drawdown Ranking

Benefits:

- Provides holistic fund evaluation.
- Balances risk and return considerations.
- Simplifies fund comparison.

Observations:

- Top-ranked funds consistently performed well across multiple metrics.
- Funds with balanced risk-return characteristics received higher scores.
- The scorecard provides a practical framework for fund selection.

Performance Analytics Summary:

Key findings include:

1. Long-term investing generated superior outcomes.
2. Risk-adjusted metrics identified higher-quality funds.
3. Alpha analysis revealed active management effectiveness.
4. Maximum Drawdown highlighted downside exposure.
5. Composite fund scoring simplified investment decision-making.

7. Advanced Analytics and Risk Metrics:

Advanced analytics was conducted to evaluate downside risk, investor behavior, portfolio concentration, and recommendation strategies.

These analyses extend beyond traditional performance metrics and provide deeper insights into fund and investor characteristics.

7.1 Historical Value at Risk (VAR):

Historical VAR was calculated at the 95% confidence level.

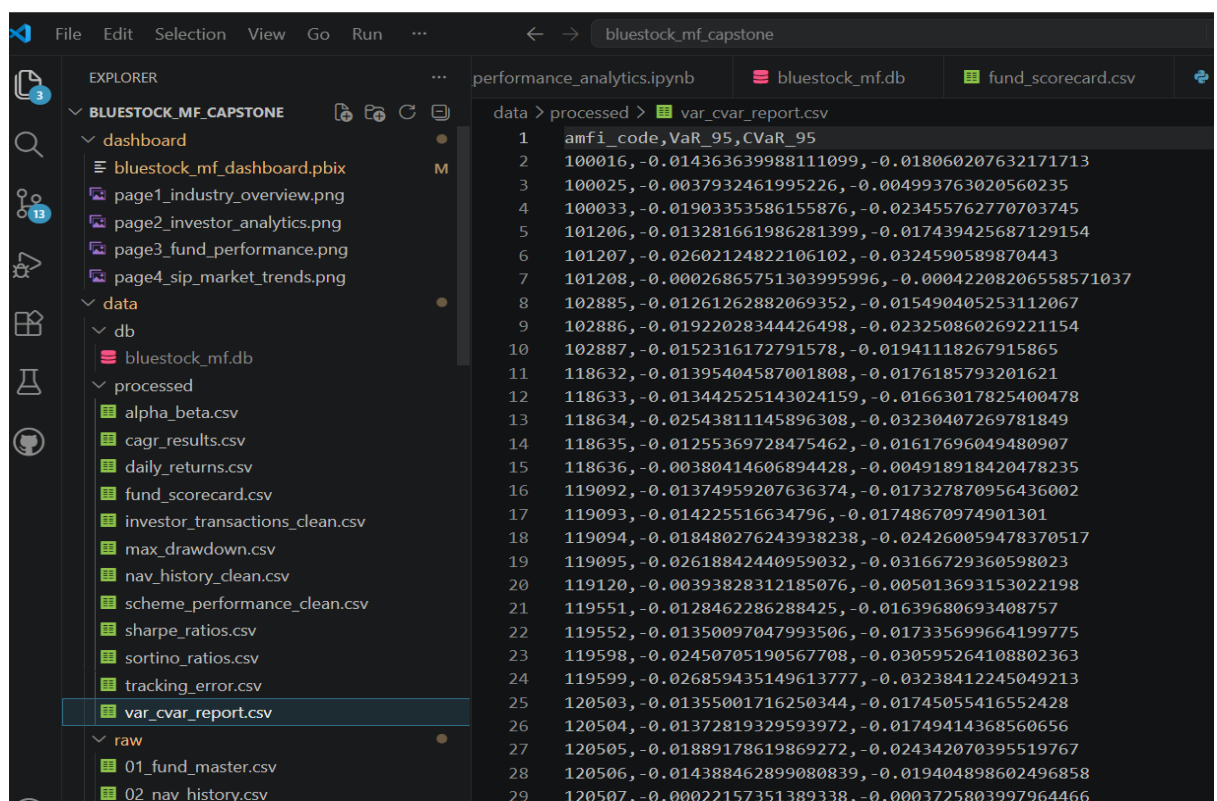
VAR estimates the maximum expected loss under normal market conditions.

Interpretation:

A VAR of -2% means that there is a 5% probability of losing more than 2% in a single day.

Observations:

- Some schemes exhibited significantly larger downside risk.
- Higher-return funds often displayed larger VAR values.
- VAR provides a useful measure of short-term risk exposure.



	amfi_code	VaR_95	CVaR_95
1	100016	-0.014363639988111099	-0.018060207632171713
2	100025	-0.0037932461995226	-0.004993763020560235
3	100033	-0.01903353586155876	-0.023455762770703745
4	101206	-0.013281661986281399	-0.017439425687129154
5	101207	-0.02602124822106102	-0.0324590589870443
6	101208	-0.00026865751303995996	-0.00042208206558571037
7	102885	-0.01261262882069352	-0.015490405253112067
8	102886	-0.01922028344426498	-0.023250860269221154
9	102887	-0.0152316172791578	-0.01941118267915865
10	118632	-0.01395404587001808	-0.0176185793201621
11	118633	-0.013442525143024159	-0.01663017825400478
12	118634	-0.02543811145896308	-0.03230407269781849
13	118635	-0.01255369728475462	-0.01617696049480907
14	118636	-0.00380414606894428	-0.004918918420478235
15	119092	-0.01374959207636374	-0.017327870956436002
16	119093	-0.014225516634796	-0.01748670974901301
17	119094	-0.018480276243938238	-0.024260059478370517
18	119095	-0.02618842440959032	-0.03166729360598023
19	119120	-0.00393828312185076	-0.005013693153022198
20	119551	-0.0128462286288425	-0.01639680693408757
21	119552	-0.01350097047993506	-0.017335699664199775
22	119598	-0.02450705190567708	-0.030595264108802363
23	119599	-0.026859435149613777	-0.03238412245049213
24	120503	-0.01355001716250344	-0.01745055416552428
25	120504	-0.01372819329593972	-0.01749414368560656
26	120505	-0.01889178619869272	-0.024342070395519767
27	120506	-0.014388462899080839	-0.019404898602496858
28	120507	-0.00022157351389338	-0.0003725803997964466
29			

7.2 Conditional Value at Risk (CVaR):

CVaR measures the average loss beyond the VAR threshold.

Observations:

- CVaR values were consistently more negative than VAR.
- Extreme losses were concentrated in high-volatility schemes.
- Tail-risk analysis is critical for risk-sensitive investors.

7.3 Rolling 90-Day Sharpe Ratio:

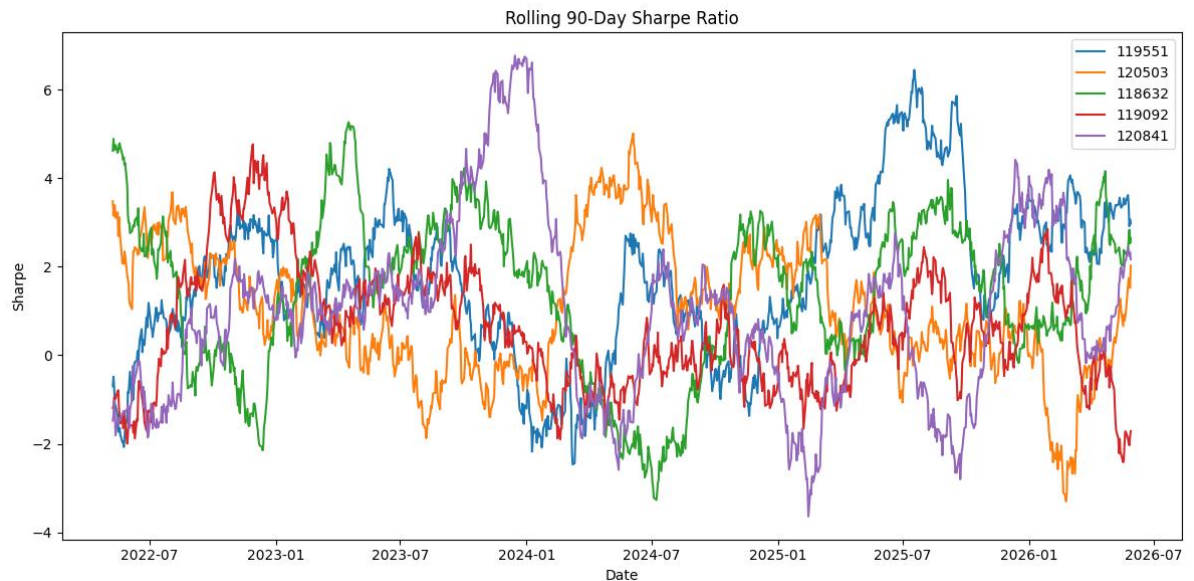
Rolling Sharpe analysis was performed for five key funds.

Purpose:

- Monitor changes in risk-adjusted performance over time.
- Identify periods of strong or weak performance.

Observations:

- Rolling Sharpe values fluctuated with market conditions.
- Some funds maintained consistently strong risk-adjusted returns.
- Performance persistence varied across schemes.



7.4 SIP Continuity Analysis:

Investor transaction histories were analysed to evaluate SIP consistency.

Investors with average gaps exceeding 35 days were classified as At-Risk.

Observations:

- Most investors maintained regular SIP contributions.
- A smaller group displayed discontinuity risk.
- Early identification enables proactive investor engagement.

7.5 Fund Recommendation System:

A rule-based fund recommendation engine was developed.

Input Parameters:

- Low Risk
- Moderate Risk
- High Risk

Recommendation Logic:

- Filter funds based on risk grade.
- Rank by Sharpe Ratio.
- Recommend top three schemes.

Benefits:

- Simple and transparent recommendation process.
- Aligns investor preferences with fund characteristics.
- Demonstrates practical application of analytics.

Example Recommendations:

Low Risk Investors

Recommended funds with stable performance and lower volatility.

Moderate Risk Investors

Recommended funds balancing growth and risk.

High Risk Investors

Recommended funds offering higher growth potential with greater volatility.

The screenshot shows a VS Code editor with the 'recommender.py' script open. The script defines a function that takes a risk level and returns a list of fund recommendations. The terminal output shows the script being executed with the risk level 'High', resulting in a list of three funds: Kotak Flexicap Fund, Mirae Asset Tax Saver Fund, and ICICI Pru Midcap Fund.

```
39         "amfi_code",
40         "scheme_name",
41         "risk_category",
42         "sharpe_ratio"
43     ]
44 ]
45
```

Problems (Ctrl+Shift+M) - Total 5 Problems

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS JUPYTER

```
PS C:\Users\s.sri gouri\Documents\bluestock_mf_capstone> cd scripts
PS C:\Users\s.sri gouri\Documents\bluestock_mf_capstone\scripts> python recommender.py
<StringArray>
['High', 'Moderate', 'Very High', 'Low', 'Moderately High']
Length: 5, dtype: str
Enter Risk Level (Low/Moderate/High): High
amfi_code      scheme_name      risk_category      sharpe_ratio
37  120843      Kotak Flexicap Fund - Regular - Growth      Moderately High      1.306744
32  148569      Mirae Asset Tax Saver Fund - Regular - Growth      High      1.234930
38  120505      ICICI Pru Midcap Fund - Regular - Growth      High      1.180101
PS C:\Users\s.sri gouri\Documents\bluestock_mf_capstone\scripts>
```

Advanced Analytics Summary:

Key findings include:

1. VaR and CVaR successfully quantified downside risk.
2. Rolling Sharpe analysis highlighted changing market conditions.
3. Investor cohorts showed increasing SIP participation.
4. Concentrated portfolios exhibited higher risk.
5. The recommender system provided practical fund-selection guidance.

8. Power BI Dashboard Development:

A four-page interactive dashboard was developed using Power BI.

The dashboard integrates cleaned datasets, performance metrics, and investor analytics into a single reporting interface.

Major features include:

- Interactive filters
- Drill-through functionality
- KPI cards
- Dynamic charts
- Fund performance comparison
- Investor analytics
- Market trend analysis

8.1 Dashboard Page 1 – Industry Overview:

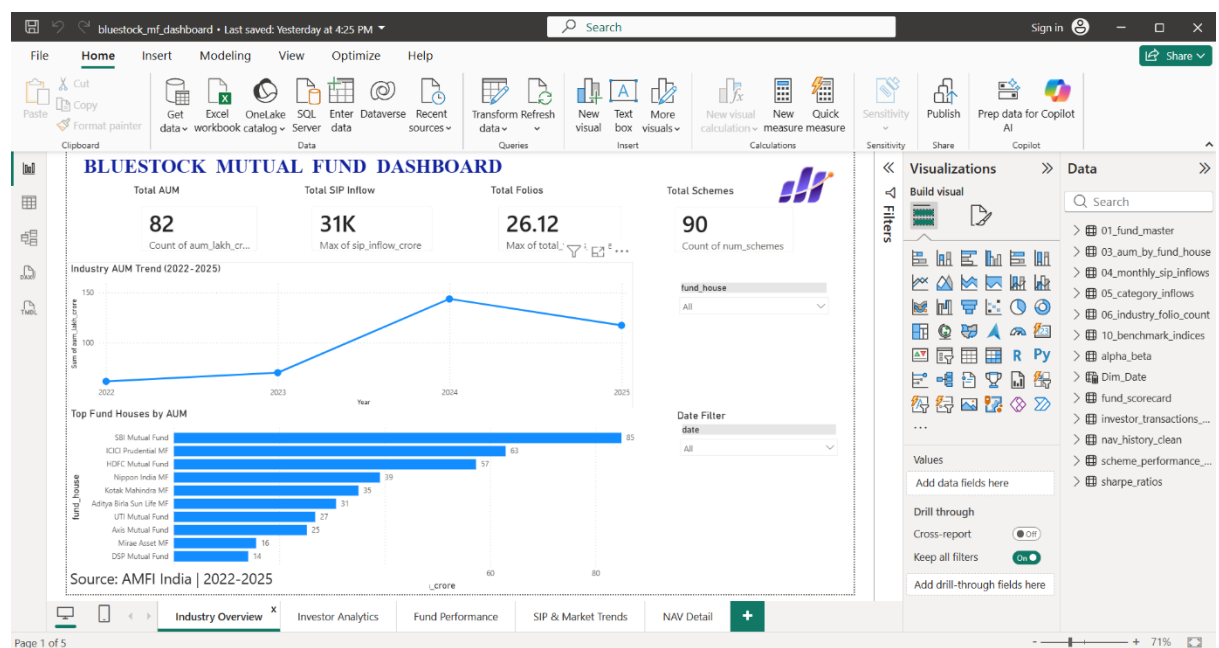
This page provides a high-level overview of the mutual fund industry.

Key Components:

- Total AUM KPI
- SIP Inflows KPI
- Total Folios KPI
- Total Schemes KPI
- Industry AUM Trend
- AUM by Fund House

Business Value:

Provides quick visibility into industry growth and market size.



8.2 Dashboard Page 2 – Investor Analytics:

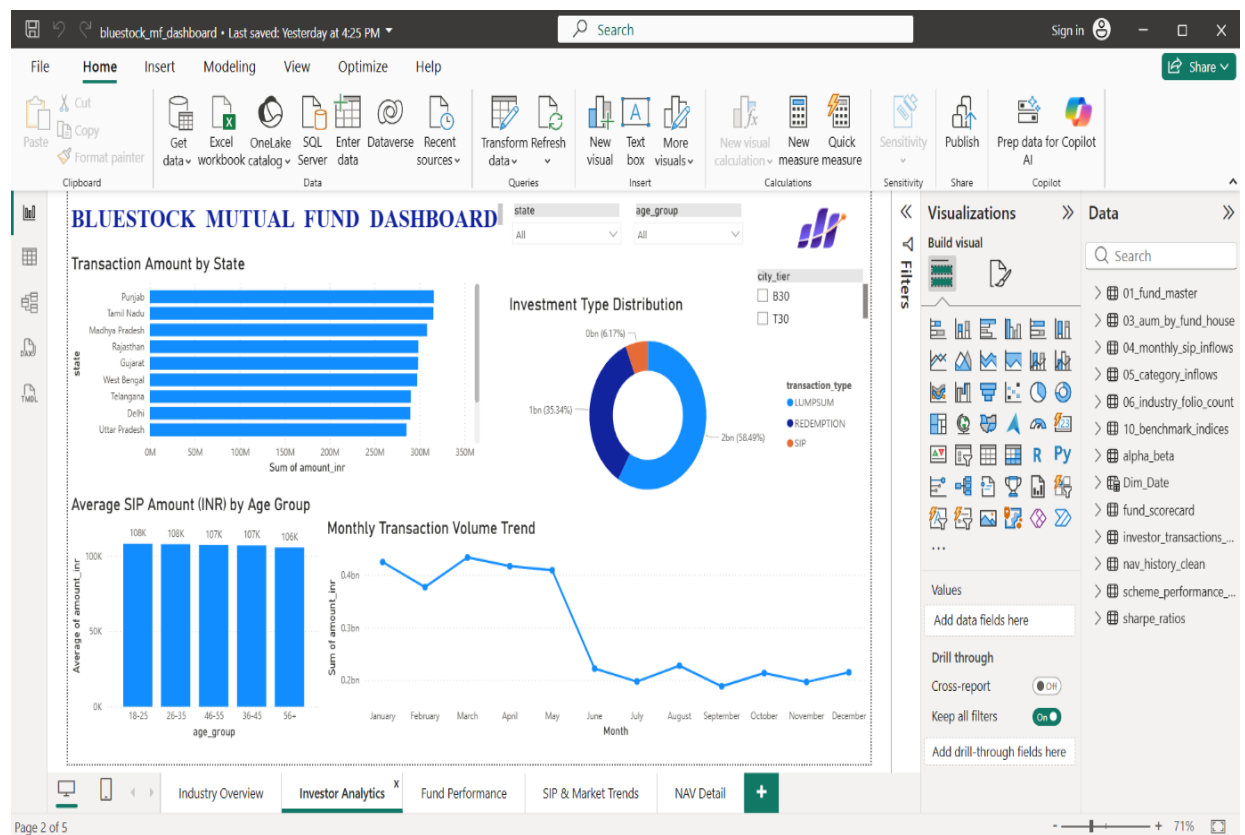
This page focuses on investor behaviour and transaction patterns.

Key Components:

- Transaction Amount by State
- SIP vs Lumpsum vs Redemption Distribution
- Age Group Analysis
- Monthly Transaction Trends

Business Value:

Helps understand investor demographics and participation patterns.



8.3 Dashboard Page 3 – Fund Performance:

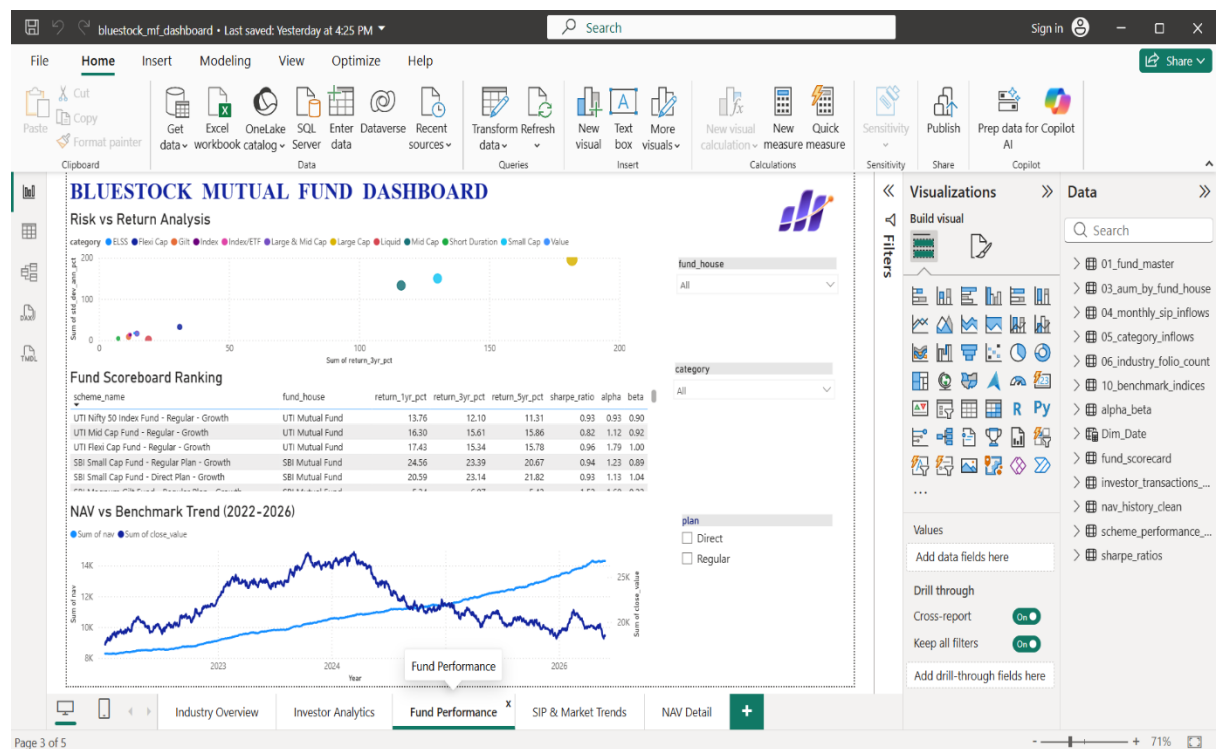
This page evaluates fund performance and risk-adjusted returns.

Key Components:

- Return vs Risk Scatter Plot
- Fund Scorecard
- Benchmark Comparison
- Fund-Level Filters

Business Value:

Supports investment evaluation and fund comparison.



8.4 Dashboard Page 4 – SIP & Market Trends:

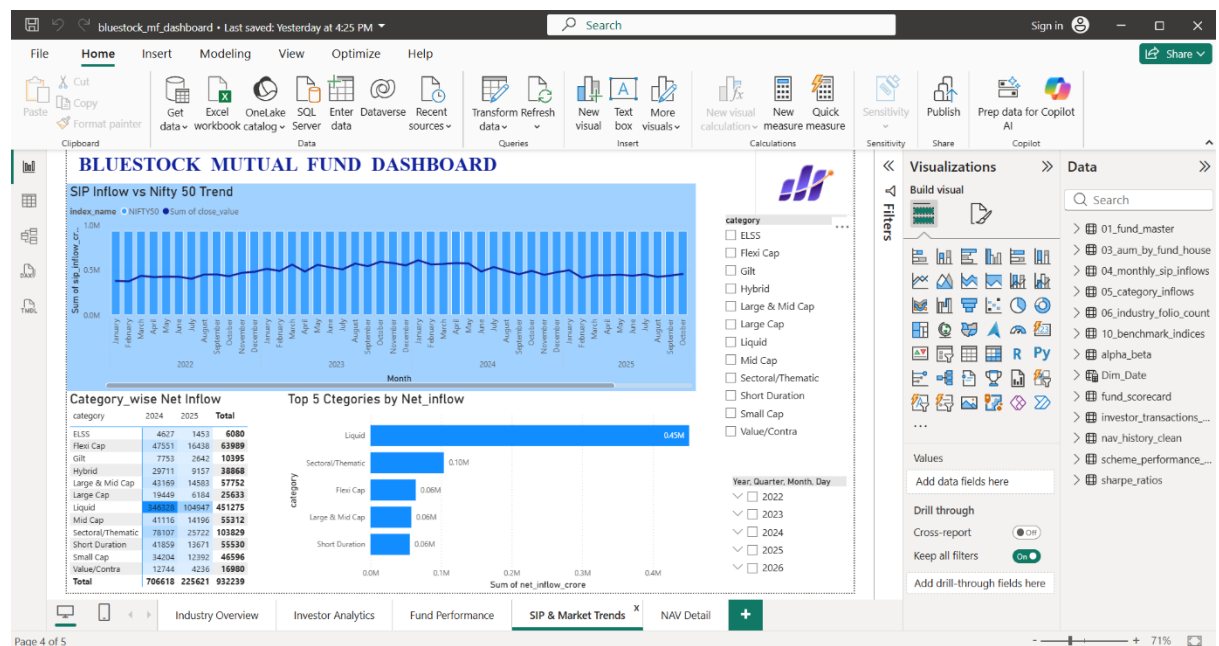
This page focuses on market participation and category inflows.

Key Components:

- SIP Growth Trends
- Nifty Comparison
- Category Inflow Analysis
- Top Performing Categories

Business Value:

Provides insights into investor sentiment and industry trends.



Dashboard Summary:

The Power BI dashboard successfully transformed analytical outputs into an interactive decision-support platform.

Benefits include:

- Improved accessibility of insights.
- Faster decision-making.
- Better investor understanding.
- Interactive exploration of mutual fund data.

9. Limitations:

Although the project achieved its objectives, certain limitations remain.

1. Historical performance does not guarantee future returns.
 2. Risk metrics rely on historical data assumptions.
 3. Portfolio holdings may change over time.
 4. Benchmark selection can influence Alpha and Beta calculations.
 5. Investor transaction data is limited to available records.
 6. The recommendation engine is rule-based and not machine-learning driven.
 7. Market events beyond the study period are not incorporated.
 8. Dashboard performance depends on data refresh frequency.
-

10. Recommendations:

Based on the analysis, the following recommendations are proposed:

For Investors

- Focus on long-term SIP investing.
- Evaluate funds using risk-adjusted metrics rather than returns alone.
- Diversify across sectors and categories.
- Monitor downside-risk indicators such as VAR and Maximum Drawdown.

For Fund Managers

- Improve portfolio diversification.
- Reduce excessive sector concentration.
- Enhance investor retention programs.
- Monitor SIP discontinuity indicators.

For Future Enhancements

- Develop machine-learning-based recommendation systems.
 - Build predictive NAV forecasting models.
 - Implement portfolio optimization techniques.
 - Automate periodic reporting.
 - Deploy dashboards to cloud platforms.
-

11. Conclusion

The Bluestock Mutual Fund Analytics Capstone Project successfully delivered a complete end-to-end analytics solution for mutual fund analysis.

The project integrated data engineering, database management, financial analytics, risk management, and business intelligence into a unified framework. Ten datasets were processed through an ETL pipeline, stored in a SQLite database, and analysed using Python-based statistical and financial techniques.

Exploratory Data Analysis revealed key trends in investor behaviour, fund performance, SIP participation, and industry growth. Advanced financial metrics including CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, VAR, and CVaR provided deeper insights into fund quality and risk exposure.

The Power BI dashboard transformed analytical findings into an interactive and user-friendly reporting platform. Additional features such as investor cohort analysis, SIP continuity monitoring, sector concentration analysis, and a fund recommendation engine enhanced the practical value of the solution.

Overall, the project demonstrates the effective application of Python, SQL, SQLite, and Power BI in solving real-world financial analytics problems and supports data-driven decision-making for investors and fund managers.

References:

1. AMFI (Association of Mutual Funds in India)
2. MFAPI Mutual Fund Data API